

2.10.51

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Question: If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three non-zero, non-coplanar vectors and

$$\begin{aligned}\mathbf{b}_1 &= \mathbf{b} - \frac{\mathbf{b} \cdot \mathbf{a}}{|\mathbf{a}|^2} \mathbf{a}, & \mathbf{b}_2 &= \mathbf{b} + \frac{\mathbf{b} \cdot \mathbf{a}}{|\mathbf{a}|^2} \mathbf{a}, \\ \mathbf{c}_1 &= \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|^2} \mathbf{a} + \frac{\mathbf{b} \cdot \mathbf{c}}{|\mathbf{c}|^2} \mathbf{b}_1, & \mathbf{c}_2 &= \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|^2} \mathbf{a} + \frac{\mathbf{b}_1 \cdot \mathbf{c}}{|\mathbf{b}_1|^2} \mathbf{b}_1, \\ \mathbf{c}_3 &= \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{c}|^2} \mathbf{a} + \frac{\mathbf{b} \cdot \mathbf{c}}{|\mathbf{c}|^2} \mathbf{b}_1, & \mathbf{c}_4 &= \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{c}|^2} \mathbf{a} = \frac{\mathbf{b} \cdot \mathbf{c}}{|\mathbf{b}|^2} \mathbf{b}_1,\end{aligned}$$

then the set of orthogonal vectors is

- 1) $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_3)$ 2) $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_2)$ 3) $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_1)$ 4) $(\mathbf{a}, \mathbf{b}_2, \mathbf{c}_2)$

solution: Given three non-zero, non-coplanar vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, define:

$$\mathbf{b}_1 = \mathbf{b} - \frac{\mathbf{b} \cdot \mathbf{a}}{|\mathbf{a}|^2} \mathbf{a} \quad (1)$$

$$\mathbf{c}_2 = \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|^2} \mathbf{a} + \frac{\mathbf{b}_1 \cdot \mathbf{c}}{|\mathbf{b}_1|^2} \mathbf{b}_1 \quad (2)$$

$$\mathbf{a} \cdot \mathbf{b}_1 = \mathbf{a} \cdot \mathbf{b} - \frac{\mathbf{b} \cdot \mathbf{a}}{|\mathbf{a}|^2} |\mathbf{a}|^2 = 0. \quad (3)$$

$\mathbf{b}_1$ is orthogonal to $\mathbf{a}$.

$$\mathbf{a} \cdot \mathbf{c}_2 = \mathbf{a} \cdot \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|^2} |\mathbf{a}|^2 + \frac{\mathbf{b}_1 \cdot \mathbf{c}}{|\mathbf{b}_1|^2} \mathbf{a} \cdot \mathbf{b}_1 \quad (4)$$

$$= \mathbf{a} \cdot \mathbf{c} - \mathbf{c} \cdot \mathbf{a} + 0 \quad (5)$$

$$= 0 \quad (6)$$

$(\mathbf{a} \cdot \mathbf{b}_1 = 0 \text{ as above})$

$$\mathbf{b}_1 \cdot \mathbf{c}_2 = \mathbf{b}_1 \cdot \mathbf{c} - \frac{\mathbf{c} \cdot \mathbf{a}}{|\mathbf{a}|^2} \mathbf{b}_1 \cdot \mathbf{a} + \frac{\mathbf{b}_1 \cdot \mathbf{c}}{|\mathbf{b}_1|^2} |\mathbf{b}_1|^2 \quad (7)$$

$$= \mathbf{b}_1 \cdot \mathbf{c} - \mathbf{b}_1 \cdot \mathbf{c} \quad (8)$$

$$= 0 \quad (9)$$

$(\mathbf{b}_1 \cdot \mathbf{a} = 0 \text{ from above})$

$\mathbf{c}_2$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}_1$

Therefore, the orthogonal set is: $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_2)$